

Part I

Optimization

Lecture Outline

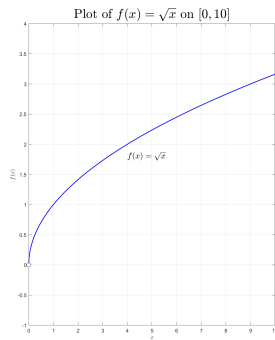
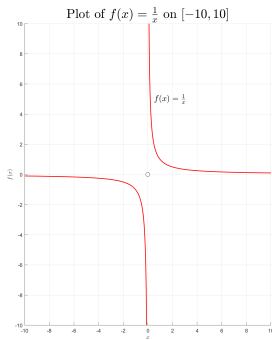
- ① The optimization problem
- ② Calculus revisited
- ③ Necessary and sufficient conditions

The optimization problem

Consider a given function

$$f : \text{dom}(f) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x)$$

with the domain $\text{dom}(f) \subseteq \mathbb{R}^n$ of f being the largest open set in $\mathbb{R}^n$ on which f can be defined. Moreover, let $R \subseteq \text{dom}(f)$ be a given set.



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In the first part of this course we study the problem of finding the points $x' \in R$ with the property that $f(x')$ is small compared to other function values $f(x)$ for x close to x' . To make this precise we introduce the notion of minimizers and minima

Minimizers and minima

Definition

Consider the function

$$f : \text{dom}(f) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}; \quad x \mapsto f(x)$$

and let $R \subseteq \text{dom}(f)$. A point $x' \in R$ is called

- A (weak) [strong] local minimizer of f if there exists $\epsilon > 0$ such that

$$(f(x) \geq f(x')) \quad [f(x) > f(x')].$$

for all $x \in R$ with $\|x - x'\| < \epsilon$ [and $x \neq x'$].

- A (weak) [strong] global minimizer of f (relative to R) if

$$(f(x) \geq f(x')) \quad [f(x) > f(x')].$$

for all $x \in R$.

The value $f(x')$ at a minimizer x' is called a minimum and are collectively called minima (or minimums).

The definition of maximizers and maxima is obtained by reversing the inequalities above. The minimizers and maximizers are collectively called extremum points, and the minima and maxima are collectively called extrema.

Minimizers and minima

Example

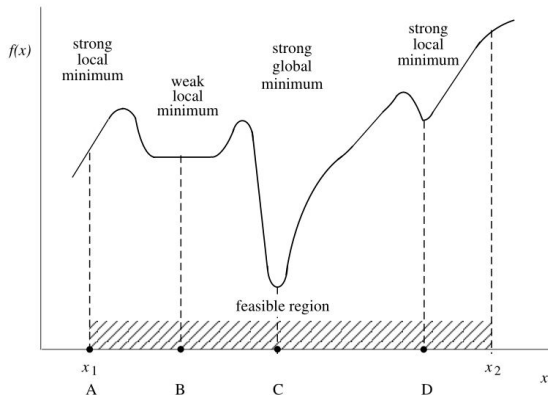


Figure 2.2. Types of minima. (Example 2.1)

The optimization problem

The problem of finding the minima of a given function

$$f : \text{dom}(f) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x)$$

relative to a given set $R \subseteq \text{dom}(f)$ is referred to as an (constrained) optimization problem and written mathematically as

$$\min_{x \in R} f(x) \tag{1}$$

- The set R is called the feasible region, and it describes the constraint of the optimization problem.
- The function f is usually referred to as the objective function or the cost function. In machine learning, the term loss function is also used.

The optimization problem

The problem of finding the minima of a given function

$$f : \text{dom}(f) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x)$$

relative to a given set $R \subseteq \text{dom}(f)$ is referred to as an (constrained) optimization problem and written mathematically as

$$\min_{x \in R} f(x) \tag{1}$$

The optimization problem (1) is sometimes written as

$$\min\{f(x) \mid x \in R\}$$

or

$$\min f(x) \text{ s.t } x \in R$$

or

$$\begin{array}{ll} \min f(x) \\ \text{s.t} \\ x \in R \end{array}$$

where s.t is short for 'subject to'.

The optimization problem

The problem of finding the minima of a given function

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relative to a given set $R \subseteq \text{dom}(f)$ is referred to as an (constrained) optimization problem and written mathematically as

$$\min_{x \in R} f(x) \tag{1}$$

When $R = \text{dom}(f)$ the optimization problem (1) is said to be unconstrained and sometimes written simply as

$$\min f(x) \quad \text{or} \quad \min_x f(x) \tag{2}$$

The optimization problem

The problem of finding the minima of a given function

$$f : \text{dom}(f) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x)$$

relative to a given set $R \subseteq \text{dom}(f)$ is referred to as an (constrained) optimization problem and written mathematically as

$$\min_{x \in R} f(x) \tag{1}$$

Note that when finding minima one typically also need to find the corresponding minimizers. The problem of find only the minimizers is also referred to as an optimization problem and written mathematically as

$$\operatorname{argmin}_{x \in R} f(x)$$

The optimization problem

The problem of finding the minima of a given function

$$f : \text{dom}(f) \subseteq \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x)$$

relative to a given set $R \subseteq \text{dom}(f)$ is referred to as an (constrained) optimization problem and written mathematically as

$$\min_{x \in R} f(x) \tag{1}$$

Note the relations

$$\begin{aligned} \min_{x \in R} f(x) &= - \max_{x \in R} -f(x) \\ \operatorname{argmin}_{x \in R} f(x) &= \operatorname{argmax}_{x \in R} -f(x) \end{aligned}$$

The optimization problem

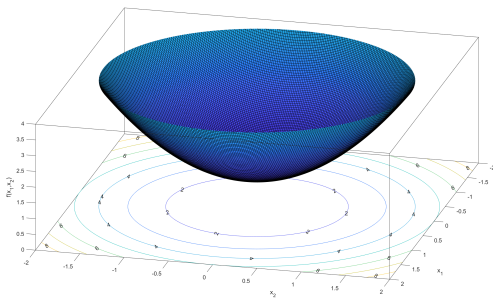
Example

Consider the optimization problem:

$$\min f(x)$$

with

$$f(x) = x_1^2 + x_2^2 + 1$$



The optimization problem

Example

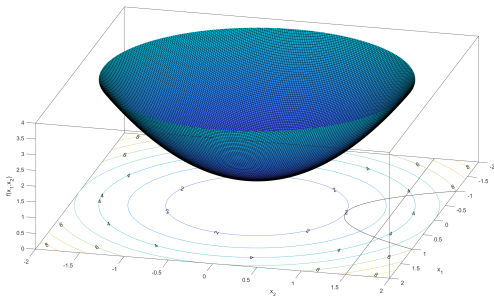
Consider the optimization problem:

$$\min_{x \in R} f(x)$$

with

$$f(x) = x_1^2 + x_2^2 + 1,$$

$$R = \{x \in \mathbb{R}^2 \mid x_2 \geq x_1^2 + 1, x_2 \leq 1.5\}$$



The optimization problem

Example

Consider the optimization problem:

$$\min_{x \in R} f(x)$$

with

$$f(x) = x_1^2 + x_2^2 + 1,$$

$$R = \{x \in \mathbb{R}^2 \mid x_2 \geq x_1^2 + 1, x_2 \leq 1.5\}$$

Note that if we write

$$c_1(x) = -x_2 + x_1^2 + 1, \text{ and } c_2(x) = -x_2 + 1.5,$$

then we may write the feasible region R as

$$R = \{x \in \mathbb{R}^2 \mid 0 \geq c_1(x), 0 \leq c_2(x)\}$$

Calculus revisited

In order to find a solution to the optimization problem:

$$\min_{x \in R} f(x)$$

that is, to find a minimizer, we need to recall some elements from calculus:

- The notion of a function f being continuous differentiable: $f \in C^1$
- The gradient ∇f and Hessian H_f of a function f
- The Jacobian matrix J_Φ of a vector valued function Φ

Continuous differentiable

Definition

Consider the function

$$f : \text{dom}(f) \subset \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x)$$

We write $f \in C^1$ and say that f is continuous differentiable if all its first order partial derivatives exists and are continuous, that is, if

$$\frac{\partial f}{\partial x_i}(x) \text{ exists for all } i = 1, \dots, n \text{ and } x \in \text{dom}(f),$$

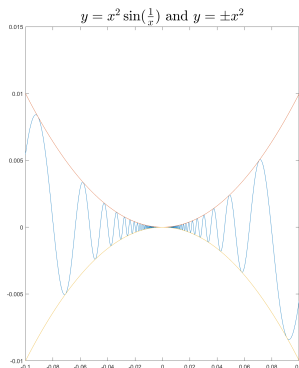
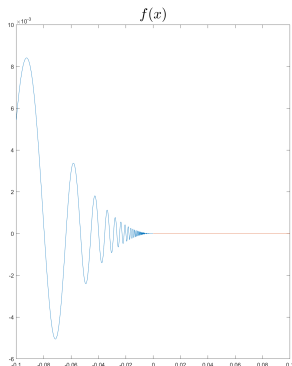
and

$$x \mapsto \frac{\partial f}{\partial x_i}(x) \text{ is continuous on } \text{dom}(f) \text{ for all } i = 1, \dots, n.$$

Continuous differentiable

Example

$$f(x) = \begin{cases} x^2 \sin(1/x), & x < 0 \\ 0, & x \geq 0 \end{cases}, \quad f'(x) = \begin{cases} 2x \sin(1/x) - \cos(1/x), & x < 0 \\ 0, & x \geq 0 \end{cases}$$

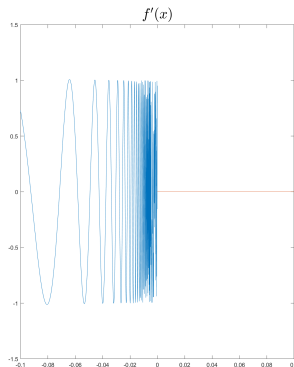
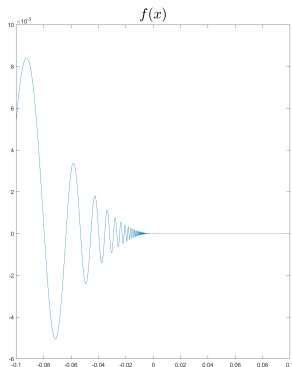


Continuous differentiable

Example

$$f(x) = \begin{cases} x^2 \sin(1/x), & x < 0 \\ 0, & x \geq 0 \end{cases}$$

is differentiable but not C^1



Continuous differentiable

Definition

Consider the function

$$f : \text{dom}(f) \subset \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x)$$

We write $f \in C^1$ and say that f is continuous differentiable if all its first order partial derivatives exists and are continuous, that is, if

$$\frac{\partial f}{\partial x_i}(x) \text{ exists for all } i = 1, \dots, n \text{ and } x \in \text{dom}(f),$$

and

$$x \mapsto \frac{\partial f}{\partial x_i}(x) \text{ is continuous on } \text{dom}(f) \text{ for all } i = 1, \dots, n.$$

We write $f \in C^2$ and say that f is twice continuous differentiable if all its first and second order partial derivatives exists and are continuous, that is, if $f \in C^1$ and

$$\frac{\partial^2 f}{\partial x_i \partial x_j}(x) \text{ exists for all } i, j = 1, \dots, n, \text{ and all } x \in \text{dom}(f),$$

and

$$x \mapsto \frac{\partial^2 f}{\partial x_i \partial x_j}(x) \text{ is continuous on } \text{dom}(f) \text{ for all } i, j = 1, \dots, n.$$

Continuous differentiable

Definition

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For vector valued functions

$$\Phi : \text{dom}(\Phi) \subset \mathbb{R}^n \rightarrow \mathbb{R}^m; x \mapsto \Phi(x) = [\Phi_1(x) \cdots \Phi_m(x)]^\top$$

with $\text{dom}(\Phi)$ open, we write $\Phi \in C^k$, $k = 1, 2$, if all the coordinate functions

$$\Phi_j : \text{dom}(\Phi) \subset \mathbb{R}^n \rightarrow \mathbb{R}, \quad j = 1, \dots, m$$

are C^k , that is, if $\Phi_j \in C^k$ for all $j = 1, \dots, m$.

Gradient, Hessian and Jacobian matrix

Definition

Consider the functions

$$f : \text{dom}(f) \subset \mathbb{R}^n \rightarrow \mathbb{R}; x \mapsto f(x)$$

$$\Phi : \text{dom}(\Phi) \subset \mathbb{R}^n \rightarrow \mathbb{R}^m; x \mapsto \Phi(x) = [\Phi_1(x) \cdots \Phi_m(x)]^\top$$

If $f \in C^1$ we define the gradient (of f at x) as the n -dimensional (column)¹ vector

$$\nabla f(x) = \left[\frac{\partial f}{\partial x_1}(x) \cdots \frac{\partial f}{\partial x_n}(x) \right]^\top \in \mathbb{R}^n$$

If $f \in C^2$ we define the Hessian (of f at x) as the $n \times n$ matrix

$$H_f(x) = \left[\frac{\partial^2 f}{\partial x_j \partial x_i}(x) \right] \in \mathbb{R}^{n \times n} \quad (2)$$

If $\Phi \in C^1$ we define the Jacobian matrix (of Φ at x) as the $m \times n$ matrix

$$J_\Phi(x) = \left[\frac{\partial \Phi_i}{\partial x_j}(x) \right] = [\nabla \Phi_1(x) \cdots \nabla \Phi_m(x)]^\top \in \mathbb{R}^{m \times n} \quad (3)$$

- The indexing in (2) and (3) indicate row i and column j .

¹Warning: some authors define the gradient as a row vector.

Gradient, Hessian and Jacobian matrix

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- For real valued functions the transpose of the gradient is just the Jacobian matrix:

$$J_f(x) = \nabla f(x)^\top$$

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Gradient, Hessian and Jacobian matrix

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- The Hessian is symmetric (since $f \in C^2$):

$$H_f(x) = H_f(x)^\top$$

¹Warning: some authors define the gradient as a row vector.

Gradient, Hessian and Jacobian matrix

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- If we let $g_i(x) = \frac{\partial f}{\partial x_i}(x)$ then

$$J_{\nabla f}(x) = [\nabla g_1(x) \cdots \nabla g_n(x)]^\top = H_f(x)$$

¹Warning: some authors define the gradient as a row vector.

Approximation via the gradient and Hessian

By continuity of ∇f and H_f we obtain for, $\|d\|$ small, the following approximations of f at x based on the gradient and Hessian:

- Linear (in d) approximation: $f(x + d) \approx f(x) + \nabla f(x)^T d$.
- Quadratic (in d) approximation: $f(x + d) \approx f(x) + \nabla f(x)^T d + \frac{1}{2} d^T H_f(x) d$.

Technical remark: The above is a consequence of the following classical result.

Theorem (Mean-value theorem for differentiation)

Let $x \in R$, $d \in \mathbb{R}^n$ and assume that

$$\{x + \alpha d \mid 0 \leq \alpha \leq \hat{\alpha}\} \subset R$$

for some $0 < \hat{\alpha} \leq 1$.

- If $f \in C^1$ then there exists $0 \leq \alpha \leq \hat{\alpha}$ such that

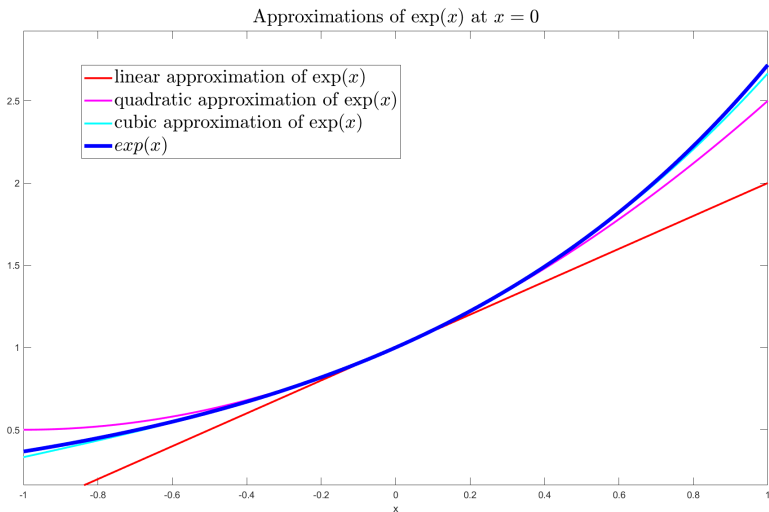
$$f(x + d) = f(x) + \nabla f(x + \alpha d)^T d.$$

- If $f \in C^2$ then there exists $0 \leq \alpha \leq \hat{\alpha}$ such that

$$f(x + d) = f(x) + \nabla f(x + \alpha d)^T d + \frac{1}{2} d^T H_f(x + \alpha d) d.$$

Approximation via the gradient and Hessian

Example



Necessary and sufficient conditions

There are two sets of conditions which are relevant for the solution of the optimization problem

$$\min_{x \in R} f(x)$$

- Necessary conditions which are satisfied at a minimizer x' .
- Sufficient conditions that guarantee that x' is a minimizer

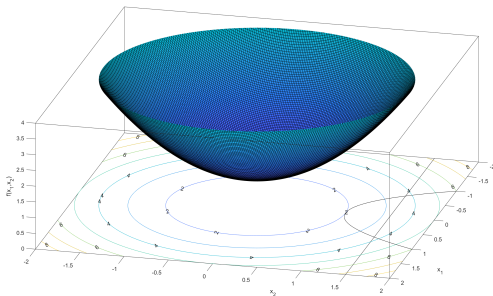
Feasible direction

Definition

A vector $d \in \mathbb{R}^n$ is called a feasible direction (wrt., R) at point $x \in \mathbb{R}^n$ if there exists $\hat{\alpha} > 0$ such that

$$x + \alpha d \in R,$$

for all $0 \leq \alpha \leq \hat{\alpha}$. The set of feasible directions at x is denoted $\mathcal{F}(x)$.



Necessary conditions

Theorem (First order necessary conditions)

Consider the optimization problem

$$\min_{x \in R} f(x)$$

with $f \in C^1$ and let $\text{int}(R)$ denote the interior of the feasible region R .

If x' is a local minimizer, then

$$\nabla f(x')^\top d \geq 0 \text{ for all } d \in \mathcal{F}(x'),$$

$$\nabla f(x') = 0 \text{ if } x' \in \text{int}(R).$$

Interior points $x' \in \text{int}(R)$ where $\nabla f(x') = 0$ are called stationary (or critical) points.

Necessary conditions

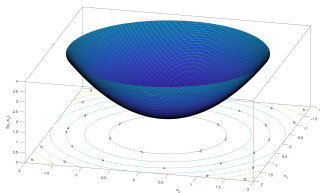
Theorem (First order necessary conditions)

Consider the optimization problem

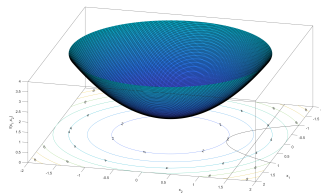
$$\min_{x \in R} f(x)$$

with $f \in C^1$ and let $\text{int}(R)$ denote the interior of the feasible region R .
If x' is a local minimizer, then

$$\begin{aligned}\nabla f(x')^\top d &\geq 0 \text{ for all } d \in \mathcal{F}(x'), \\ \nabla f(x') &= 0 \text{ if } x' \in \text{int}(R).\end{aligned}$$



(a)



(b)

Necessary conditions

Theorem (Second order necessary conditions)

Consider the optimization problem

$$\min_{x \in R} f(x)$$

with $f \in C^2$ and let $\text{int}(R)$ denote the interior of the feasible region R .
If x' is a local minimizer, then

- For all $d \in \mathcal{F}(x')$

$$\nabla f(x')^\top d \geq 0,$$

$$\text{If } \nabla f(x')^\top d = 0 \text{ then } d^\top H_f(x')d \geq 0.$$

- If $x' \in \text{int}(R)$

$$\nabla f(x') = 0,$$

$$d^\top H_f(x')d \geq 0 \text{ for all } d \neq 0.$$

Hence, for a stationary point x' to be a local minimizers, the Hessian $H_f(x')$ at x' must be positive semidefinite.

Necessary conditions

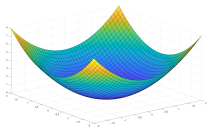
Example

See if $x' = (0, 0)$ is a local minimizer for

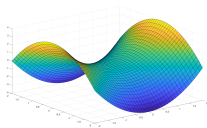
$$f_1(x) = x_1^2 + x_2^2, \quad R = \mathbb{R}^2,$$

$$f_2(x) = x_1^2 - x_2^2, \quad R = \mathbb{R}^2,$$

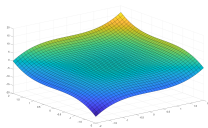
$$f_3(x) = x_1^3 + x_2^3, \quad R = \mathbb{R}^2.$$



(c) Graph of f_1



(d) Graph of f_2



(e) Graph of f_3

Necessary conditions

Example

See if $x' = (0, 0)$ is a local minimizer for

$$f_1(x) = x_1^2 + x_2^2, \quad R = \mathbb{R}^2,$$

$$f_2(x) = x_1^2 - x_2^2, \quad R = \mathbb{R}^2,$$

$$f_3(x) = x_1^3 + x_2^3, \quad R = \mathbb{R}^2.$$

We compute

$$\nabla f_1(x') = (0, 0), \quad H_{f_1}(x') = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix},$$

$$\nabla f_2(x') = (0, 0), \quad H_{f_2}(x') = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix},$$

$$\nabla f_3(x') = (0, 0), \quad H_{f_3}(x') = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Hence x' is a local minimizer **candidate** for f_1 and f_3 , but not for f_2 .

Note that x' would be a minimizer **candidate** for f_2 if e.g., R is the cone spanned by $(1, 1)$ and $(1, -1)$.

Sufficient conditions

Theorem (Second order sufficient conditions)

Consider the optimization problem

$$\min_{x \in R} f(x)$$

with $f \in C^2$ and let $\text{int}(R)$ denote the interior of the feasible region R .

If $x' \in \text{int}(R)$ and

$$\nabla f(x') = 0,$$

$$d^\top H(x') d > 0 \text{ for all } d \neq 0,$$

then x' is a strong local minimizer.

Hence, if the Hessian $H_f(x')$ is positive definite at a stationary point x' , then it's a strong local minimizer.

Sufficient conditions

Example (revised)

See if $x' = (0, 0)$ is a local minimizer for

$$f_1(x) = x_1^2 + x_2^2, \quad R = \mathbb{R}^2,$$

$$f_2(x) = x_1^2 - x_2^2, \quad R = \mathbb{R}^2,$$

$$f_3(x) = x_1^3 + x_2^3, \quad R = \mathbb{R}^2.$$

We compute

$$\nabla f_1(x') = (0, 0), \quad H_{f_1}(x') = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix},$$

$$\nabla f_2(x') = (0, 0), \quad H_{f_2}(x') = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix},$$

$$\nabla f_3(x') = (0, 0), \quad H_{f_3}(x') = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Hence x' is a strong local minimizer for f_1 .

Note that we can't say anything conclusive in the case of f_3 .